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(57) Abstract:

An AI-based automatic visual inspection system (100) for MSME manufacturing comprising a transparent inspection enclosure (102), a part-handling robotic arm (104) with gripper end-effector (116), a first camera robotic arm (106) and second camera robotic arm (108) each carrying high-resolution industrial cameras (110, 112), a diffused LED illumination module (118) with strobe trigger controller (120), and an edge computing platform (122) executing an AI defect detection engine (124). The part-handling arm (104) picks, positions, and reorients the part inside the transparent enclosure (102) while the camera arms (106, 108) capture multi-view images from complementary angular positions. The AI defect detection engine (124) performs deep-learning inference on the multi-view image set to detect and classify surface defects including scratches, cracks, burrs, dimensional deviations, and pits, returning a pass/fail decision with defect location and confidence score. The handling arm (104) sorts the part to accept (128) or reject (130) bins and the data logging module (132) records full inspection traceability metadata. The system achieves inspection cycle times of 8 to 15 seconds, detection accuracy exceeding 92%, and is deployable on low-cost edge hardware making it accessible for small and medium manufacturing enterprises.

FORM 2

THE PATENTS ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. TITLE OF THE INVENTION

**AI-BASED AUTOMATIC VISUAL INSPECTION SYSTEM FOR SMALL
SCALE MSME MANUFACTURING**

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3. PREAMBLE TO THE DESCRIPTION The following specification particularly describes the invention and the manner in which it is to be performed.		

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FIELD

The present disclosure relates to the field of automated quality control and manufacturing inspection technologies. More particularly, the present disclosure relates to an AI-based automatic visual inspection system for small scale MSME manufacturing units, comprising dual robotic camera arms, a part-handling robotic arm, a transparent inspection enclosure, and an AI-powered defect detection engine, and a method thereof.

DEFINITIONS

15

For the purpose of this specification, the following terms shall have the meanings assigned to them below. These definitions are provided to clarify terminology used throughout the description and claims and are not intended to limit the scope of the invention.

"AI-Based Visual Inspection System" refers to an integrated automated quality control apparatus comprising robotic arms, image-acquisition cameras, a transparent inspection enclosure, illumination subsystem, and an artificial intelligence engine configured to detect surface and dimensional defects on manufactured parts without human intervention.

"Handling Robotic Arm" refers to a multi-degree-of-freedom articulated robot fitted with an end-effector gripper, configured to pick a part from a loading station, position and reorient it inside the transparent inspection enclosure, and transfer it to an accept or reject output bin based on the AI decision.

"Camera Robotic Arm" refers to an articulated robotic arm carrying an industrial camera and lens assembly at its end-effector, capable of moving to pre-programmed angular and positional coordinates to capture images of the inspected part from multiple viewpoints.

"Transparent Inspection Enclosure" refers to a fully transparent acrylic or polycarbonate box structure that houses the part during inspection, providing a controlled, shadow-free optical environment while allowing cameras positioned on all sides to acquire unobstructed images.

"Defect Detection AI Engine" refers to a deep-learning-based computer vision software module,

5 running on an embedded edge computing platform, that receives multi-view images of a part, performs inference using a trained neural network model, and classifies the part as conforming or non-conforming, identifying defect type, location, and severity.

"Multi-View Image Set" refers to the collection of digital images captured by the camera arms from different angular positions around the transparent inspection enclosure in a single inspection cycle, providing complete surface coverage of the part under inspection.

"Edge Computing Platform" refers to a compact embedded computing device such as NVIDIA Jetson Nano or Raspberry Pi with AI accelerator, deployed locally within the inspection cell to execute AI inference without dependence on cloud connectivity.

"End-Effector" refers to the tool mounted at the tip of a robotic arm, which may be a two-finger or three-finger pneumatic or servo-actuated gripper for the handling arm, or a camera-and-lens mount for the camera arms.

"Strobe Trigger" refers to a synchronisation signal that simultaneously activates the LED illumination flash and the camera shutter to freeze part motion and eliminate motion blur during image acquisition.

"Defect Classification" refers to the assignment of a detected anomaly to a predefined category such as scratch, crack, burr, dimensional deviation, surface pit, or discolouration, based on learned visual features.

BACKGROUND

25 The background information herein below relates to the present disclosure but is not necessarily prior art.

Small and medium manufacturing enterprises (MSMEs) face persistent challenges in maintaining consistent part quality due to reliance on manual visual inspection. Human inspectors are subject to fatigue, inconsistency, and subjective judgment, resulting in high rates of defect escape to customers and excessive false-rejection of conforming parts. Manual inspection is also a bottleneck that limits production throughput.

Existing automated optical inspection systems are primarily designed for flat surfaces in electronics and printed circuit board manufacturing and are not directly applicable to three-dimensional machined or cast components. These systems require expensive gantry setups, controlled conveyor integration, and dedicated optical benches that are economically prohibitive for MSME-scale operations.

Robotic inspection systems available in the market typically require single-arm configurations with limited angular coverage, resulting in blind spots on complex-geometry parts. Without a controlled illumination enclosure, reflections and ambient light variation cause false positives and reduce detection accuracy.

Conventional machine vision systems rely on rule-based image processing algorithms that require extensive manual programming for each new part geometry. These systems lack the adaptability of deep learning approaches and demand significant engineering effort for every product changeover.

The integration of multiple articulated robotic arms with a transparent enclosure for full-surround inspection, combined with an AI-based defect detection engine deployable on low-cost edge hardware, addresses these limitations. Such a system enables MSMEs to achieve automated inspection accuracy comparable to large-scale manufacturing lines at a fraction of the capital cost.

Given these factors, there is a need for an AI-based automatic visual inspection system suitable for MSME manufacturing that provides complete three-dimensional surface coverage, AI-driven defect

5 classification, and low-cost deployable edge computing inference.

OBJECTS

10 Some of the objects of the present disclosure, which at least one embodiment herein satisfies, are as follows:

It is an object of the present disclosure to ameliorate one or more problems of the prior art or to at least provide a useful alternative.

15 An object of the present disclosure is to provide an AI-based automatic visual inspection system for MSME manufacturing that achieves complete three-dimensional surface coverage of parts using dual camera robotic arms and a transparent inspection enclosure.

Another object of the present disclosure is to provide a part handling robotic arm that autonomously picks, orients, and repositions the part during inspection to expose all surfaces to the camera arms without manual handling.

20 Still another object of the present disclosure is to provide a deep-learning-based defect detection engine capable of identifying surface defects including scratches, cracks, burrs, pits, and dimensional deviations with accuracy exceeding 92%.

Yet another object of the present disclosure is to provide a transparent inspection enclosure with integrated diffused LED illumination that ensures shadow-free, uniform lighting on all surfaces of the part under inspection.

25 Still another object of the present disclosure is to provide an inspection system deployable on low-cost edge computing hardware, making the technology economically accessible to small and medium manufacturing enterprises.

Yet another object of the present disclosure is to provide an inspection cycle time of 8 to 15 seconds per part, enabling integration with existing MSME production lines without throughput reduction.

30 Still another object of the present disclosure is to provide a data logging and traceability module that records inspection images, defect classifications, and pass/fail decisions with timestamps and part identifiers for quality audit purposes.

35 Yet another object of the present disclosure is to provide a modular and scalable system architecture that supports operation with one camera arm initially and upgrade to dual camera arms as production volume increases.

Other objects and advantages of the present disclosure will be more apparent from the following description, which is not intended to limit the scope of the present disclosure.

SUMMARY

40 The present disclosure envisages an AI-based automatic visual inspection system for MSME manufacturing. The system comprises a transparent inspection enclosure, a part-handling robotic arm, two camera robotic arms, a multi-camera image acquisition subsystem, a diffused LED illumination module, an edge computing platform executing an AI defect detection engine, and a sorting and data logging module.

The transparent inspection enclosure is a fully transparent acrylic box that houses the part during inspection and provides optical access to the part from all directions simultaneously.

The part-handling robotic arm is a multi-DOF articulated arm with a gripper end-effector configured

5 to pick parts from a loading station, place them in the transparent inspection enclosure, reorient them during the inspection cycle, and transfer them to accept or reject bins based on the AI decision. The two camera robotic arms are articulated arms each carrying a high-resolution industrial camera with macro or telecentric lens, positioned at complementary angular coordinates around the transparent inspection enclosure to capture multi-view images without blind spots.

10 The diffused LED illumination module is integrated within or around the transparent inspection enclosure and provides uniform, shadow-free lighting on all surfaces of the part, synchronised with camera capture via strobe trigger.

The edge computing platform is an embedded device executing the AI defect detection engine locally within the inspection cell, receiving multi-view image sets and returning pass/fail decisions with defect classification in real time.

15 The AI defect detection engine is a deep-learning convolutional neural network model trained on part-specific defect datasets, performing inference on multi-view image sets to detect and classify surface defects including scratches, cracks, burrs, dimensional deviations, surface pits, and discolouration, returning a pass/fail decision with defect location and confidence score.

20 The sorting and data logging module receives the AI decision and actuates the handling arm to route the part to the appropriate output bin, while recording inspection metadata including timestamp, part identifier, images, defect type, and confidence score.

In an embodiment, the AI defect detection engine is based on YOLOv8 or EfficientDet architecture trained using transfer learning from pretrained COCO weights with part-specific fine-tuning on 500 to 2000 images per defect class.

25 In an embodiment, the two camera arms capture 4 to 8 images per inspection cycle from positions at 0 degrees, 45 degrees, 90 degrees, and oblique angles relative to the part, providing complete surface coverage.

In an embodiment, the edge computing platform is an NVIDIA Jetson Nano or Raspberry Pi with Google Coral Edge TPU accelerator, providing AI inference latency below 500 milliseconds per image.

30 In an embodiment, the transparent inspection enclosure includes access ports for the handling arm and camera arms and is lined with diffused LED ring lights or backlit panels to eliminate specular reflections.

In an embodiment, the system supports a modular upgrade path allowing initial deployment with one camera arm and upgrade to dual camera arms as production requirements increase.

35 In an embodiment, the data logging module stores inspection records on a local server and optionally synchronises with a cloud-based quality management dashboard for OEE reporting and customer audit trail generation.

The present disclosure further envisages a method for AI-based automatic visual inspection of manufactured parts. The method comprises the following steps:

40

- loading a part onto the loading station, wherein the handling robotic arm picks the part using its gripper end-effector and places it within the transparent inspection enclosure;
- positioning the camera robotic arms at first pre-programmed angular coordinates around the transparent inspection enclosure;
- 45 • activating the diffused LED illumination module via strobe trigger and simultaneously triggering image capture by the camera arms to acquire a first subset of multi-view images;
- commanding the handling robotic arm to reorient the part to expose previously occluded surfaces;
- repositioning the camera robotic arms at second pre-programmed angular coordinates and

- 5 capturing a second subset of multi-view images to complete the multi-view image set;
- transmitting the multi-view image set to the edge computing platform executing the AI defect detection engine;
- performing AI inference on the multi-view image set to detect, localise, and classify surface defects, returning a pass or fail decision with defect type, location, and confidence score;
- 10 • actuating the handling robotic arm to transfer the part to the accept bin on pass decision or to the reject bin on fail decision; and
- recording inspection metadata including timestamp, part identifier, captured images, defect classification, confidence score, and final decision in the data logging module.

15 **BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS**

An AI-based automatic visual inspection system for MSME manufacturing, and a method thereof, will now be described with the help of the accompanying drawings, in which:

20 Figure 1 illustrates a system architecture overview of the AI-based automatic visual inspection system showing the transparent inspection enclosure, dual camera robotic arms, part-handling robotic arm, edge computing platform, and data logging module, in accordance with an embodiment of the present disclosure;

Figure 2 illustrates a top-view schematic of the inspection cell layout showing the spatial arrangement of the transparent enclosure, camera arm positions, handling arm workspace, loading station, and 25 accept/reject bins, in accordance with an embodiment of the present disclosure; and

Figure 3 illustrates a process flow diagram of the AI-based inspection method showing the sequential steps from part loading through AI inference to sorting and data logging, in accordance with an embodiment of the present disclosure.

30 **LIST OF REFERENCE NUMERALS**

- 100 – AI-Based Visual Inspection System
- 102 – Transparent Inspection Enclosure
- 104 – Part-Handling Robotic Arm
- 35 106 – First Camera Robotic Arm
- 108 – Second Camera Robotic Arm
- 110 – Industrial Camera (First)
- 112 – Industrial Camera (Second)
- 114 – Macro / Telecentric Lens
- 40 116 – Gripper End-Effector
- 118 – Diffused LED Illumination Module
- 120 – Strobe Trigger Controller
- 122 – Edge Computing Platform
- 124 – AI Defect Detection Engine
- 45 126 – Part Loading Station
- 128 – Accept Output Bin
- 130 – Reject Output Bin
- 132 – Data Logging Module

- 5 134 – PLC / Microcontroller
- 136 – Part (Under Inspection)
- 200 – Inspection Process Flow
- 202 – Part Loading & Gripper Pick
- 204 – Camera Arm Positioning (Set 1)
- 10 206 – Strobe Trigger & Image Capture (Set 1)
- 208 – Part Reorientation by Handling Arm
- 210 – Camera Arm Positioning (Set 2)
- 212 – Strobe Trigger & Image Capture (Set 2)
- 214 – AI Inference - Defect Detection
- 15 216 – Pass / Fail Decision
- 218 – Part Sorting - Accept or Reject Bin
- 220 – Data Logging & Traceability Record

DETAILED DESCRIPTION

20

Embodiments of the present disclosure will now be described with reference to the accompanying drawings.

Embodiments are provided so as to thoroughly and fully convey the scope of the present disclosure to the person skilled in the art. Numerous details are set forth, relating to specific components and methods, to provide a complete understanding of embodiments of the present disclosure. It will be
25 apparent to the person skilled in the art that the details provided in the embodiments should not be construed to limit the scope of the present disclosure. In some embodiments, well-known processes, well-known apparatus structures, and well-known techniques are not described in detail.

The terminology used in the present disclosure is only for the purpose of explaining a particular
30 embodiment and such terminology shall not be considered to limit the scope of the present disclosure. As used in the present disclosure, the forms "a," "an," and "the" may be intended to include the plural forms as well, unless the context suggests otherwise.

In current MSME manufacturing practice, visual inspection of machined and cast parts is performed manually by trained operators, which introduces subjectivity, fatigue-induced error, and throughput
35 bottlenecks. Conventional automated inspection systems are cost-prohibitive and geometrically limited for three-dimensional part inspection. Deep learning-based visual inspection on edge computing hardware addresses these limitations at MSME-accessible cost.

To address these limitations, an AI-based automatic visual inspection system (hereinafter referred to as "system 100") and a corresponding method (hereinafter referred to as "method 200") are now
40 described with reference to Figures 1 through 3.

Referring to Figure 1, the AI-based automatic visual inspection system 100 comprises a transparent inspection enclosure 102, a part-handling robotic arm 104, a first camera robotic arm 106, a second camera robotic arm 108, a diffused LED illumination module 118, a strobe trigger controller 120, an edge computing platform 122 executing an AI defect detection engine 124, a part loading station 126,
45 an accept output bin 128, a reject output bin 130, a data logging module 132, and a PLC/microcontroller 134 for orchestration.

The transparent inspection enclosure 102 is a fully transparent acrylic box with internal dimensions sufficient to house the largest part to be inspected, typically 300 mm x 300 mm x 300 mm for general

5 MSME use, with access ports on two faces for the handling arm 104 and ventilation slots. The enclosure provides an optically controlled environment eliminating ambient light variation and uncontrolled shadows.

The part-handling robotic arm 104 is a 4 to 6 degree-of-freedom articulated arm fitted with a two-finger or three-finger pneumatic or servo gripper end-effector 116. The arm 104 operates under G-code or
10 ROS-based motion control generated by the PLC/microcontroller 134, with workspace radius sufficient to reach the loading station 126, the interior of the enclosure 102, and both output bins 128, 130.

The first camera robotic arm 106 and second camera robotic arm 108 are articulated arms each carrying a high-resolution industrial camera 110, 112 with a macro or telecentric lens 114. The arms move to pre-programmed angular positions during the inspection cycle, providing top-view, side-view, oblique-
15 view, and bottom-view image acquisition. Each camera has a resolution of 12 to 48 megapixels with strobe synchronisation via the trigger controller 120.

The diffused LED illumination module 118 comprises LED ring lights or backlit panels integrated within or immediately surrounding the transparent enclosure 102. Diffusion panels eliminate specular reflections from metallic part surfaces. The illumination is synchronised with camera capture via the
20 strobe trigger controller 120, producing short-duration flash exposures that freeze residual part motion. In an embodiment, the diffused LED illumination module 118 provides coaxial lighting for flat surfaces and low-angle dark-field lighting for surface crack detection, selectable by the PLC/microcontroller 134 based on part material and defect target profile.

The edge computing platform 122 is an embedded computing device, such as NVIDIA Jetson Nano or
25 Raspberry Pi 4 with Google Coral Edge TPU accelerator, deployed locally within the inspection cell. The platform 122 receives multi-view image sets from the cameras 110, 112 via USB 3.0 or GigE interface, executes the AI defect detection engine 124, and returns inference results within 500 milliseconds per image.

The AI defect detection engine 124 is a convolutional neural network model based on YOLOv8 or
30 EfficientDet architecture, trained using transfer learning and fine-tuned on a dataset of 500 to 2000 labelled images per defect class. The engine 124 detects and classifies defects including scratches, cracks, burrs, dimensional deviations, surface pits, and discolouration, returning bounding box coordinates, defect class, and confidence score for each detected anomaly.

In an embodiment, the AI defect detection engine 124 is periodically retrained using inspection images
35 accumulated in the data logging module 132, enabling continuous learning and accuracy improvement as production data grows.

In an embodiment, the AI defect detection engine 124 supports multi-class simultaneous defect detection, enabling identification of multiple defect types on a single part in a single inference pass.

Referring to Figure 2, the top-view schematic of the inspection cell illustrates the spatial arrangement
40 of the transparent enclosure 102 at the centre, the two camera arms 106, 108 positioned on opposite sides, the handling arm 104 at the front access port, the loading station 126 to the left, and the accept bin 128 and reject bin 130 to the right of the enclosure.

Referring to Figure 3, the inspection process flow 200 begins at step 202 with the part being picked from the loading station 126 by the handling arm 104 and placed inside the transparent inspection
45 enclosure 102. At step 204, the camera arms 106, 108 move to their first set of pre-programmed angular coordinates. At step 206, the strobe trigger controller 120 fires the LED illumination and simultaneously triggers image capture on both cameras 110, 112, acquiring the first subset of multi-view images. At step 208, the handling arm 104 reorients the part to expose previously occluded

5 surfaces. At steps 210 and 212, the camera arms reposition and capture the second image subset. At step 214, the complete multi-view image set is transmitted to the edge computing platform 122 and the AI defect detection engine 124 performs inference. At step 216, the pass/fail decision is generated. At step 218, the handling arm 104 transfers the part to the accept bin 128 or reject bin 130. At step 220, inspection metadata is recorded in the data logging module 132.

10 The complete inspection cycle from part pick to sort completion operates within 8 to 15 seconds per part, depending on part complexity, number of image captures, and AI inference time on the deployed edge platform.

15 In an embodiment, the system 100 is connected to a cloud-based quality dashboard aggregating inspection data from multiple inspection cells, providing real-time OEE metrics, defect trend analysis, and customer-facing quality audit reports.

In an embodiment, the system 100 supports part changeover by loading a new AI model checkpoint and updated camera arm position program specific to the new part geometry, enabling multi-product operation within a single inspection cell.

20 Safety monitoring of the system 100 includes collision detection on all robotic arms via torque sensing or proximity sensors, emergency stop integration with the PLC/microcontroller 134, and fail-safe procedures that halt all arm motion on detection of unexpected obstruction or sensor anomaly.

TECHNICAL ADVANCEMENTS

25 The present disclosure described herein above has several technical advantages including, but not limited to, the realization of an AI-based automatic visual inspection system and method that:

- achieves complete three-dimensional surface coverage of manufactured parts through the coordinated action of dual camera robotic arms and a fully transparent inspection enclosure, eliminating blind spots inherent in single-camera or fixed-camera systems;
- 30 • provides AI-based defect detection accuracy exceeding 92%, reducing defect escape rate and false rejection rate compared to manual inspection, which typically achieves 70 to 80% accuracy under sustained production conditions;
- enables inspection cycle times of 8 to 15 seconds per part, supporting integration with existing MSME production lines without throughput reduction;
- 35 • deploys on low-cost edge computing hardware making automated inspection economically accessible to small and medium manufacturing enterprises;
- integrates a transparent acrylic inspection enclosure with diffused LED illumination to provide a controlled optical environment that eliminates ambient light interference and specular reflections from metallic part surfaces;
- 40 • supports transfer learning-based AI model training requiring only 500 to 2000 images per defect class, reducing data collection burden for MSME operators;
- provides full inspection traceability through timestamped logging of inspection images, defect classifications, confidence scores, and pass/fail decisions, supporting customer quality audits and ISO certification requirements;
- 45 • supports modular scalability, allowing initial deployment with one camera arm and upgrade to dual camera arms as production volume increases, protecting the initial capital investment; and
- enables multi-product operation through rapid part changeover by loading part-specific AI model checkpoints and camera arm position programs.

5 The embodiments herein and the various features and advantageous details thereof are explained with reference to the non-limiting embodiments in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein.

10 The foregoing description of the specific embodiments so fully reveals the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments.

15 The use of the expression "at least" or "at least one" suggests the use of one or more elements or ingredients or quantities, as the use may be in the embodiment of the disclosure to achieve one or more of the desired objects or results.

20 While considerable emphasis has been placed herein on the components and component parts of the preferred embodiments, it will be appreciated that many embodiments can be made and that many changes can be made in the preferred embodiments without departing from the principles of the disclosure.

5 **WE CLAIM:**

CLAIM 1. An AI-based automatic visual inspection system (100) for MSME manufacturing, the system (100) comprising:

- 10 a transparent inspection enclosure (102) configured to house a part under inspection and provide optical access to the part from all directions;
- a part-handling robotic arm (104) fitted with a gripper end-effector (116) and configured to pick the part from a loading station (126), position and reorient the part within the transparent inspection enclosure (102), and transfer the part to an accept output bin (128) or reject output bin (130) based on an inspection decision;
- 15 a first camera robotic arm (106) and a second camera robotic arm (108), each carrying an industrial camera (110, 112) with a lens (114), positioned at complementary angular locations around the transparent inspection enclosure (102) and configured to capture multi-view images of the part from a plurality of angular positions;
- 20 a diffused LED illumination module (118) integrated with the transparent inspection enclosure (102) and synchronised with image capture via a strobe trigger controller (120);
- an edge computing platform (122) operatively connected to the cameras (110, 112) and configured to execute an AI defect detection engine (124); and
- 25 a data logging module (132) configured to record inspection metadata including timestamp, part identifier, captured images, defect classification, and pass/fail decision,
- wherein the AI defect detection engine (124) receives the multi-view images and performs inference to detect, localise, and classify surface defects on the part, returning a pass/fail decision.

30 CLAIM 2. The AI-based automatic visual inspection system (100) as claimed in claim 1, wherein the AI defect detection engine (124) is based on a convolutional neural network architecture trained using transfer learning from pretrained weights and fine-tuned on a part-specific defect dataset of 500 to 2000 images per defect class.

35 CLAIM 3. The AI-based automatic visual inspection system (100) as claimed in claim 1, wherein the AI defect detection engine (124) detects and classifies defects including scratches, cracks, burrs, dimensional deviations, surface pits, and discolouration, returning bounding box coordinates, defect class, and confidence score for each detected anomaly.

40 CLAIM 4. The AI-based automatic visual inspection system (100) as claimed in claim 1, wherein the first camera robotic arm (106) and second camera robotic arm (108) capture 4 to 8 images per inspection cycle from angular positions including 0 degrees, 45 degrees, 90 degrees, and oblique positions relative to the part.

45 CLAIM 5. The AI-based automatic visual inspection system (100) as claimed in claim 1, wherein the transparent inspection enclosure (102) is constructed of clear acrylic or polycarbonate material with access ports for the part-handling robotic arm (104) and the camera robotic arms (106, 108).

CLAIM 6. The AI-based automatic visual inspection system (100) as claimed in claim 1, wherein the

5 edge computing platform (122) is an NVIDIA Jetson Nano or Raspberry Pi with an AI accelerator providing inference latency below 500 milliseconds per image.

CLAIM 7. The AI-based automatic visual inspection system (100) as claimed in claim 1, wherein the
10 diffused LED illumination module (118) provides selectable coaxial illumination for flat surface inspection and low-angle dark-field illumination for surface crack detection.

CLAIM 8. The AI-based automatic visual inspection system (100) as claimed in claim 1, wherein the
15 system (100) further comprises a cloud-based quality dashboard configured to aggregate inspection data, display real-time OEE metrics, defect trend analysis, and generate customer-facing quality audit reports.

CLAIM 9. The AI-based automatic visual inspection system (100) as claimed in claim 1, wherein the
20 system (100) supports multi-product operation through rapid changeover by loading part-specific AI model checkpoints and camera arm position programs for each new part geometry.

CLAIM 10. A method for AI-based automatic visual inspection of manufactured parts in an MSME
setting, the method comprising:

picking a part from a loading station (126) using a gripper end-effector (116) of a part-handling
robotic arm (104) and placing the part within a transparent inspection enclosure (102);

25 positioning a first camera robotic arm (106) and a second camera robotic arm (108) at first pre-programmed angular coordinates around the transparent inspection enclosure (102);

activating a diffused LED illumination module (118) via a strobe trigger controller (120) and
simultaneously triggering image capture on industrial cameras (110, 112) carried by the
camera robotic arms (106, 108) to acquire a first subset of multi-view images;

30 commanding the part-handling robotic arm (104) to reorient the part to expose previously occluded surfaces;

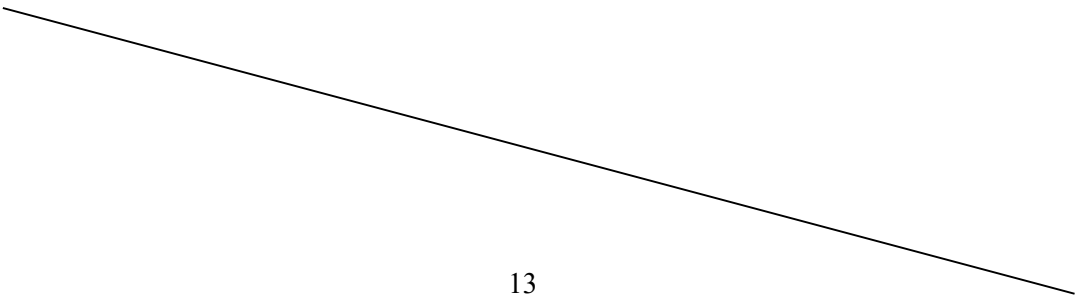
repositioning the camera robotic arms (106, 108) at second pre-programmed angular
coordinates and capturing a second subset of multi-view images to complete a multi-view
image set;

35 transmitting the multi-view image set to an edge computing platform (122) executing an AI defect detection engine (124);

performing AI inference on the multi-view image set to detect, localise, and classify surface
defects, returning a pass or fail decision with defect type, location, and confidence score;

40 actuating the part-handling robotic arm (104) to transfer the part to an accept output bin (128) on a pass decision or to a reject output bin (130) on a fail decision; and

recording inspection metadata including timestamp, part identifier, captured images, defect
classification, confidence score, and final decision in a data logging module (132).



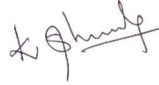
5 **Dated this 04th day of May, 2026**



RAJKUMAR,
IMMANUAL



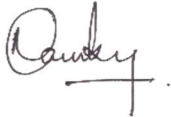
NITHYANANTHA
KUMAR,
SANJANA



SHEELA SOBANA
RANI



ALAN
SAHAYARAJ



NAVEENKUMAR
BALASUBRAMAN
IYAN



SENBAGAPRIYA
SELVAN



KAMALESHWARAN
PERUMAL



KALAISELVAN
SRINIVASAN



PRASANNA
THANGAPANDI

10 **TO,**
15 **THE CONTROLLER OF PATENTS**
20 **THE PATENT OFFICE, AT CHENNAI**

AI-BASED AUTOMATIC VISUAL INSPECTION SYSTEM FOR SMALL SCALE MSME MANUFACTURING

10 An AI-based automatic visual inspection system (100) for MSME manufacturing comprising a transparent inspection enclosure (102), a part-handling robotic arm (104) with gripper end-effector (116), a first camera robotic arm (106) and second camera robotic arm (108) each carrying high-resolution industrial cameras (110, 112), a diffused LED illumination module (118) with strobe trigger controller (120), and an edge computing platform (122) executing an AI defect detection engine (124). The part-handling arm (104) picks, positions, and reorients
15 the part inside the transparent enclosure (102) while the camera arms (106, 108) capture multi-view images from complementary angular positions. The AI defect detection engine (124) performs deep-learning inference on the multi-view image set to detect and classify surface defects including scratches, cracks, burrs, dimensional deviations, and pits, returning a pass/fail decision with defect location and confidence score. The handling arm (104) sorts the part to
20 accept (128) or reject (130) bins and the data logging module (132) records full inspection traceability metadata. The system achieves inspection cycle times of 8 to 15 seconds, detection accuracy exceeding 92%, and is deployable on low-cost edge hardware making it accessible for small and medium manufacturing enterprises.